

Elastic Net Regression: From First Principles

Elastic Net combines Ridge and Lasso. Ridge shrinks coefficients smoothly using an L2 penalty, while Lasso can push some coefficients exactly to zero using an L1 penalty. Elastic Net is useful when we want both stability and feature selection.

Why Elastic Net Is Needed

Lasso can be unstable when features are strongly correlated because it may choose one feature and ignore another similar feature. Ridge handles correlated features more smoothly but does not remove weak features. Elastic Net sits between them.

Core Formula

Elastic Net keeps the Linear Regression prediction equation but changes the cost function.

$$J(\theta) = MSE + \lambda_1 \sum_{j=1}^n |\theta_j| + \lambda_2 \sum_{j=1}^n \theta_j^2$$

The L1 part can remove features. The L2 part stabilizes the coefficient values.

L1 Ratio

Many libraries use an `l1_ratio` value. If it is close to 1, Elastic Net behaves more like Lasso. If it is close to 0, it behaves more like Ridge.

How It Learns

Elastic Net searches for coefficients that reduce prediction error while also keeping the model simple. It rewards useful features, shrinks weak effects, and may remove features that do not help enough.

When to Use It

Use Elastic Net when there are many features, correlated features, and possible irrelevant features. Avoid it when the relationship is highly non-linear and a linear model is too simple.